Lecture 11 : Project Evaluation and Programme Management

Introduction

In the dynamic world of software development, projects are not merely technical endeavors but significant investments that demand careful scrutiny and strategic oversight. Just as a business evaluates any potential investment, software projects must undergo rigorous assessment to ensure they align with organizational goals, offer tangible returns, and are viable from multiple perspectives. This lecture delves into the critical processes of project evaluation and program management, providing a framework for making informed decisions about which projects to pursue, how to manage a collection of projects effectively, and how to assess individual project viability. Understanding these concepts is crucial for aspiring project managers and stakeholders, as they directly impact resource allocation, risk mitigation, and ultimately, the success and profitability of an organization's software initiatives.

Core Concepts

Term	Definition	Significance
`Business Case`	A comprehensive document, often synonymous with a feasibility study, that justifies the initiation of a project by demonstrating its financial and technical viability.	Provides the foundational rationale for a project, ensuring benefits outweigh costs and risks are considered.
`Feasibility Study`	An analysis that assesses the practicality of a proposed project, focusing on technical, financial, and operational aspects.	Determines if a project is achievable and worthwhile before significant resources are committed.
`Technical Feasibility`	Evaluation of whether a project's required functionality can be achieved using currently available and affordable technologies.	Ensures the project is technologically possible and avoids embarking on unachievable technical goals.

Term	Definition	Significance
`Financial Feasibility`	Assessment of a project's economic viability, primarily by comparing its anticipated benefits against its estimated costs.	Guarantees that the project is a sound financial investment, yielding sufficient returns.
`Operational Feasibility`	Determination of whether a proposed system will be accepted and effectively utilized by its intended end-users and integrated into existing operations.	Crucial for user adoption and successful integration, preventing resistance to change.
`Project Portfolio Management (PPM)`	A centralized approach to managing a collection of projects, providing an overview, prioritizing resources, and making strategic decisions about project initiation and termination.	Optimizes resource allocation and strategic alignment across an organization's entire project landscape.
`Project Risk`	Threats or uncertainties that could negatively impact the successful execution and completion of a project.	Requires proactive identification and mitigation to ensure project delivery.
`Business Risk`	Factors that threaten the anticipated benefits or value of a delivered project, even if the project is technically completed.	Focuses on the long-term value and impact of the project on the organization's objectives.
`Cost-Benefit Analysis`	A systematic process of identifying, quantifying, and comparing the total costs of a project with its total benefits to determine its overall value.	Essential for financial assessment, ensuring that the economic advantages justify the investment.

Detailed Analysis

1. The Business Case and Feasibility Studies

A **business case** serves as the fundamental justification for undertaking a project. It is essentially a **feasibility study** that aims to determine if developing a particular software or system is financially and technically viable. The core objective is to demonstrate that the project's benefits will significantly outweigh its development, implementation, and operational costs. Furthermore, a robust business case must account for potential business risks, which are factors that could threaten the expected benefits of the delivered project.

The feasibility study typically encompasses three critical dimensions:

- **Technical Feasibility**: This assesses whether the project's required functionality can be achieved with the current, affordable technologies available to the organization. It also considers if any special hardware or software is needed and if it's readily available or likely to become available. If the necessary technology is absent or inaccessible, the project is deemed technically infeasible.
- **Financial Feasibility**: This evaluates the economic viability of the project. The primary goal is to ensure that the profits or benefits gained from implementing the project are substantially greater than the costs incurred during its development and operation. A project is financially feasible only if its benefits clearly outweigh its costs.
- **Operational Feasibility**: Even if a project is technically and financially sound, its success hinges on user acceptance. Operational feasibility examines how well the proposed system will be accepted by end-users and integrated into existing workflows. Resistance from users, perhaps due to fear of job displacement or disruption of established manual processes, can render an otherwise viable project operationally infeasible. A project should only be undertaken if it satisfies all three feasibility criteria.

2. Contents of a Business Case

A comprehensive business case or feasibility study report typically includes the following sections:

- **Introduction/Background**: This section sets the stage by describing the problem the project aims to solve or the opportunity it seeks to exploit.
- **The Proposed Project**: A brief outline of the project's scope, detailing what the project intends to achieve.

- **The Market**: If the project involves developing a new product (e.g., a computer game), this section assesses the likely demand for the product in the market.
- **Organizational and Operational Infrastructure**: This describes how the organization might need to change to accommodate the new project. This is particularly important when introducing new information systems that could alter existing structures or operations.
- **Benefits**: These should ideally be expressed in financial or monetary terms. The client, who is funding the project, will ultimately assess these benefits against the costs.
- **Outline Implementation Plan**: This details how the project will be implemented, considering any potential disruption to the organization's normal operations.
- **Costs**: This section provides a detailed breakdown of all anticipated costs, which the implementation plan helps to establish.
- **The Financial Case (Financial Analysis)**: This combines the cost and benefit data to establish the overall value of the project, determining its economic attractiveness.
- **Risks**: This section identifies potential risks. Two main types are considered:
- **Project Risks**: Threats to the successful execution and completion of the project itself.
- **Business Risks**: Factors that could threaten the benefits expected from the delivered project. The business case primarily focuses on business risks.
- **Management Plan**: A detailed plan outlining the activities and milestones for the smooth management of the project and the organization during its development.

3. Project Portfolio Management (PPM)

Project Portfolio Management (PPM) provides a holistic view of all projects an organization is undertaking, whether they are academic, healthcare, business, or industry-oriented. Its primary functions include:

- **Prioritizing Resource Allocation**: Since resources are often limited, PPM helps decide which projects are most critical and should receive resources first.
- **Project Selection and Termination**: It guides decisions on which new projects to accept based on their potential benefits and which existing projects to discontinue if they are no longer viable or are incurring losses.

The key concerns addressed by PPM include:

- **Evaluating Proposals**: Establishing criteria and processes for assessing new project proposals.
- **Assessing Risk**: Understanding the overall risk profile across all projects, including both project and business risks.
- **Resource Sharing**: Optimizing the allocation of limited resources across multiple projects.
- **Managing Dependencies**: Identifying and accounting for interdependencies between projects to ensure smooth execution.
- **Removing Duplication**: Identifying and eliminating redundant efforts or components across similar projects to save time and resources.
- **Ensuring Necessary Developments**: Making sure that critical developments are not inadvertently overlooked or missed.

PPM is structured around three main elements:

1. **Project Portfolio Definition**:
2. Creating a central record of all projects within the organization.
3. Deciding whether to include all projects or only specific categories (e.g., ICT projects).
4. Distinguishing between new product development (NPD) projects and renewal projects, as they have different resource and effort requirements.
5. **Project Portfolio Management**:
6. Recording and assessing the actual costing and performance of ongoing projects.
7. This element focuses on monitoring and controlling the portfolio.
8. **Project Portfolio Optimization**:
9. Using the gathered information to achieve a better balance across the project portfolio.
10. For example, balancing high-risk, potentially high-value projects with less risky, but less valuable projects to manage overall organizational risk and return.

Pros and Cons of PPM:
* **Pros**: Allows for small, ad-hoc tasks (like quick system fixes) to be handled outside the formal portfolio, providing flexibility.
* **Cons**: Full-time staff allocated to projects might effectively work part-time due to routine tasks, impacting efficiency. Also, if some projects are excluded from the official portfolio, it may not accurately reflect the organization's true activity.

4. Evaluation of Individual Projects

Evaluating individual projects involves a detailed assessment across technical, financial, and operational dimensions, as highlighted in the feasibility study.

- **Technical Assessment**: This involves verifying if the required functionality can be achieved using current, affordable technologies. It also considers the cost implications of adopting new technologies, hardware, or software, which must be factored into the financial assessment.
- **Financial Assessment (Cost-Benefit Analysis)**: This is a crucial step for individual projects. It requires:
- **Identifying all costs**: This includes development costs (e.g., staff salaries), set-up costs (e.g., hardware, recruitment, staff training), and operational costs (e.g., system maintenance after installation).
- **Identifying the value of benefits**: This involves quantifying the advantages gained from the project.
- **Checking that benefits are greater than costs**: The fundamental principle is that a project should only proceed if its anticipated benefits significantly outweigh its total costs. If this condition is not met, the project should not be undertaken.

Key Takeaways

- [ ] A business case, or feasibility study, is essential for justifying a project by proving its financial, technical, and operational viability.
- [ ] Projects must be assessed for technical feasibility (can it be built?), financial feasibility (is it worth building?), and operational feasibility (will it be used?).
- [ ] The benefits of a project must always exceed its costs, including development, implementation, and operational expenses.
- [ ] A comprehensive business case includes details on the project's background, scope, market, infrastructure needs, benefits, costs, risks, and management plan.
- [ ] Project Portfolio Management (PPM) provides a strategic overview of all organizational projects, aiding in resource allocation, project selection, and risk management.
- [ ] PPM involves defining the project portfolio, managing its ongoing performance, and optimizing it for balance between risk and value.
- [ ] Individual project evaluation requires a detailed cost-benefit analysis, identifying all types of costs (development, setup, operational) and benefits.

- [ ] Business risks, which threaten the benefits of a delivered project, are a primary focus in the business case.

Conclusion

The concepts of project evaluation and program management are indispensable for any organization engaged in software development. They transform project initiation from a speculative endeavor into a strategic investment decision, grounded in thorough analysis. By meticulously crafting a business case, conducting comprehensive feasibility studies, and implementing robust project portfolio management, organizations can ensure that their resources are allocated efficiently, risks are proactively managed, and projects are aligned with overarching business objectives. This systematic approach not only minimizes the chances of project failure but also maximizes the potential for delivering tangible value and fostering sustainable growth. As technology continues to evolve, the ability to critically evaluate and strategically manage a portfolio of projects will remain a cornerstone of successful software engineering and organizational leadership.